Nom i cognoms:	**Grup:**
Activitat: Exercicis de derivades	**Data:**

Exercicis de derivades

1. $f(x) = 4x^6 - 5x^5 + x^4 - 3x^3 + 10x^2 + 8x - 3$
2. $f(x) = -6e^x - 5\ln x - \sqrt{x} + 4\cos x - 3\sin x + 42$
3. $f(x) = 7\sin x + \frac{x^6}{3} + 10 - \frac{\cos x}{7} + e^x$
4. $f(x) = \frac{x^2-8x+1}{5}$
5. $f(x) = 32\sqrt[8]{x^5} + 9\sqrt[3]{x}$
6. $f(x) = e^x + 9\sin x$
7. $f(x) = 8x^4 - x^3 + 8x^2 - x + 10$
8. $f(x) = \ln x - x + 8e^x - \frac{x}{4}$
9. $f(x) = x - \cos x$
10. $f(x) = 7x + x^7 + \frac{x}{7} + \frac{7}{x}$
11. $f(x) = 12\cos x - 8\ln x$
12. $f(x) = e^x + e\,x + x^2 + e + \pi x + 3\pi$
13. $f(x) = \sqrt[3]{x} + \sqrt{x^3} + 3\sqrt{x} + \sqrt{3}\,x$
14. $f(x) = (2x^3)^7$
15. $f(x) = \frac{5}{2}x^3$
16. $f(x) = \frac{3x^4}{6}$
17. $f(x) = \frac{3}{5x^3}$
18. $f(x) = \frac{7x^3}{6} + \frac{5x^2}{8} + \frac{2x}{5} + \frac{1}{7}$
19. $f(x) = \sqrt{5}\,x^4 + \frac{1}{x}$
20. $f(x) = \sqrt{x} + x^3 \cdot x$
21. $f(x) = x^3 \cdot x^4 + x^3\sqrt{x}$
22. $f(x) = \sin(3) + e^4 + \sqrt{\pi} - 2$
23. $f(x) = \frac{(\sqrt[5]{x})^3}{\sqrt[3]{x^2}}$
24. $f(x) = x^2 \cdot \sqrt[3]{x^{12}}$
25. $f(x) = \frac{e^x+e^{-x}}{2}$

Solucions

1. $f'(x) = 24x^5 - 25x^4 + 4x^3 - 9x^2 + 20x + 8$
2. $f'(x) = -6e^x - \frac{5}{x} - \frac{1}{2\sqrt{x}} - 4\sin x - 3\cos x$
3. $f'(x) = 7\cos x + 2x^5 + \frac{\sin x}{7} + e^x$
4. $f'(x) = \frac{2x-8}{5}$
5. $f'(x) = \frac{20}{\sqrt[8]{x^3}} + \frac{3}{\sqrt[3]{x^2}}$
6. $f'(x) = e^x + 9\cos x$
7. $f'(x) = 32x^3 - 3x^2 + 16x - 1$
8. $f'(x) = \frac{1}{x} - \frac{5}{4} + 8e^x$
9. $f'(x) = 1 + \sin x$
10. $f'(x) = \frac{50}{7} + 7x^6 - \frac{7}{x^2}$
11. $f'(x) = -12\sin x - \frac{8}{x}$
12. $f'(x) = e^x + e + 2x + \pi$
13. $f'(x) = \frac{1}{3\sqrt[3]{x^2}} + \frac{3\sqrt{x}}{2} + \frac{3}{2\sqrt{x}} + \sqrt{3}$
14. $f'(x) = 2688x^{20}$
15. $f'(x) = \frac{15}{2}x^2$
16. $f'(x) = 2x^3$
17. $f'(x) = -\frac{9}{5x^4}$
18. $f'(x) = \frac{7x^2}{2} + \frac{5x}{4} + \frac{2}{5}$
19. $f'(x) = 4\sqrt{5}\,x^3 - \frac{1}{x^2}$
20. $f'(x) = \frac{1}{2\sqrt{x}} + 4x^3$
21. $f'(x) = 7x^6 + \frac{7\sqrt{x^5}}{2}$
22. $f'(x) = 0$
23. $f'(x) = -\frac{1}{15x\sqrt[15]{x}}$
24. $f'(x) = 6x^5$
25. $f'(x) = \frac{e^x - e^{-x}}{2}$